

Supplementary Information for From Trace to Modal Closure: Dynamic Conditions for Regenerative Reentry in a Path-Dependent Adaptive System

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1 Supplementary methods

1.1 Evidence hierarchy and prospective design freezing

Paper IV followed a diagnostic-to-confirmatory workflow. Diagnostic experiments were used to refine hypotheses, identify observables, or falsify candidate mechanisms. Confirmatory designs were subsequently frozen in the internal research record before fresh seed cohorts were executed. Because this freezing was not independently timestamped in an external registry, the paper uses the phrase *prospectively frozen confirmatory criteria* rather than external preregistration.

Most central confirmatory experiments used 64 fresh seeds. Important exceptions include the 32-instance common-map diagnostic and the independent 16-seed native-autocorrection replication. Inference was experiment specific and used combinations of nonparametric 95% bootstrap intervals, favorable-seed or favorable-cell fractions, one-sided sign-flip/randomization tests, Holm correction for specified hypothesis families, and frozen effect-size thresholds. Final transcript-coded analyses frequently used 150,000 bootstrap resamples, but this number is not retroactively assigned to earlier experiments where the exact analysis implementation was not independently archived. No single global multiplicity correction was imposed across the full longitudinal sequence of distinct experiments; multiplicity control was applied within the hypothesis families specified by each frozen design. The separation of fresh confirmatory cohorts from diagnostic/exploratory blocks and the complete evidence audit are therefore part of the inferential record.

1.2 Full reactor constants and dimensions

Table 1: Core reactor dimensions and constants used in Paper IV.

Quantity	Value / shape	Construction
Recurrent state x	32	4 modules $\times$ 8 units
Body/environment/action	4 each	$s, e, a \in \mathbb{R}^4$
Context c	8	$c = S_e e + S_a a + \eta_c$
Slow memory m	8	decay 0.98, write rate 0.02
Plastic state θ	4	clipped to $[-0.25, 0.25]$
Relevance r	12	decay 0.98, rate 0.02
Boundary trace K	8×4	decay 0.98, rate 0.02
History g_h	up to 5×4	exponentially weighted module means
W_0 sparsity	$p_{\text{in}} = 0.35, p_{\text{out}} = 0.05$	nonzero $\mathcal{N}(0, 0.35)$
$\rho(W_0)$	0.75	spectral normalization
B_{in}	32×12	$\mathcal{N}(0, 0.18)$
M	32×8	$\mathcal{N}(0, 0.12)$
P_m	8×32	$\mathcal{N}(0, 1/\sqrt{32})$
S_e	8×4	$\mathcal{N}(0, 0.18)$
S_a	8×4	4 nonzero rows, $\mathcal{N}(0, 0.30)$
J_p, J_r, J_b	$32 \times 4, 32 \times 12, 32 \times 8$	SD 0.04, 0.025, 0.025
Recurrent leak	0.25 nonlinear update	$x_{t+1} = 0.75x_t + 0.25 \tanh z_t$
Body noise SD	0.005	matched where required
Context noise SD	0.02	matched where required
Action exploration SD	0.02	matched where required
Finite-difference ϵ	10^{-4}	operator experiments
Write / return windows	24 / 24 steps	isolated relay experiments

The full recurrent update is given in the main Methods. For completeness, the adaptive updates that are easiest to lose when the model is described only verbally are

$$e_{t+1} = 0.95e_t + 0.03\eta_t^{\text{env}} + \mu_t, \quad (1)$$

$$r_{t+1} = 0.98r_t + 0.02u_{t-1}\Delta\mathcal{V}_t, \quad (2)$$

$$m_{t+1} = 0.98m_t + 0.02P_mx_{t+1}, \quad (3)$$

$$K_{t+1} = 0.98K_t + 0.02c_t\zeta_{t-1}^T, \quad (4)$$

$$\theta_{t+1} = \text{clip}\left[\theta_t + 0.002(\chi^* - \chi_{t+1}) - 0.004\theta_t + 0.0005\Delta\mathcal{V}_tp_t^{\text{post}}, -0.25, 0.25\right], \quad (5)$$

where $\Delta\mathcal{V}_t$ is computed from the pre-action viability at the beginning of the step, χ_{t+1} is the vector of module-wise mean squared activities after the recurrent update, and p_t^{post} includes the newly appended module-mean vector. The recurrent drive instead uses p_t^{pre} , computed from the history deque before appending the current module means.

Exact C4 within-step order. The recovered base source updates a timestep in the following order. This order is part of the model specification rather than an implementation convenience.

1. Update the environmental state e and apply any protocol-defined body kick.
2. Compute current context $c = S_e e + S_a a + \eta_c$, concatenate $u = [s; c]$, evaluate pre-action viability, and form $\Delta\mathcal{V}$ relative to the stored previous viability.
3. Update relevance r from the previous input u_{t-1} and $\Delta\mathcal{V}$.
4. Form the relevance-modulated effective input u_{eff} , construct $W(\theta)$, and initialize the recurrent drive.

5. Update K from current context and previous action noise; compute the row-norm scores, collective median, and b .
6. Compute p^{pre} from the existing history deque and add the $J_p p + J_r r + J_b b$ recurrent drives.
7. Update recurrent state x and apply any protocol-defined lesion/reset.
8. Compute current module means g and append g to the five-entry history deque.
9. Update the action from the new x -derived g , current body state, and current action noise.
10. Update body state s ; Paper-IV relation histories use $R_4(\phi)$ in the action-to-body term.
11. Update slow memory m from the new recurrent state.
12. Recompute p^{post} from the updated history deque and update θ .
13. Store current u , action noise, and the pre-action viability as the lagged quantities for the next step.

The reference reconstruction of the relational branch uses the same standard counter-clockwise planar rotation in coordinates (1, 2) and (3, 4) and is independently checked by the recovered causal-angle estimator.

Metric definitions. For nonzero vectors a, b , sign-invariant directional similarity and angle are

$$c_{\pm}(a, b) = \frac{|a^T b|}{\|a\| \|b\|}, \quad \alpha_{\pm}(a, b) = \arccos c_{\pm}(a, b). \quad (6)$$

For unit vectors v, w , one-dimensional subspace displacement is $d_P(v, w) = \|vv^T - ww^T\|_F$. The half-channel interface efficiency is

$$\eta_{\text{int}} = \frac{\sigma_1(BA)}{\sigma_1(A)\sigma_1(B)}, \quad (7)$$

and normalized non-normality is

$$\nu(T) = \frac{\|T^T T - T T^T\|_F}{\|T\|_F^2}. \quad (8)$$

Representational-similarity analysis (RSA) compares vectorized pairwise dissimilarity structures under the correlation rule frozen for that experiment. In T01, the empirical structure consists of pairwise standardized future-response distances and the target structure consists of true circular chord distances between donor relations. T06 analogously compares relative history-present angular geometry. Where the recovered design records only “correlation” rather than the exact correlation-family implementation, the release does not retroactively relabel it Pearson or Spearman. Shuffle advantages are the observed RSA minus the mean RSA under the corresponding frozen label or component shuffle.

For C07, the observed test-context closure need is

$$n_{\text{actual}} = 1 - |u_B^T v_B|, \quad (9)$$

while the dynamic prediction from the calibration map is

$$n_{\text{pred}} = 1 - |u_B^T C_{\text{dyn}}^A u_B|, \quad (10)$$

with unit-normalized modal vectors; closure-need error is $|n_{\text{pred}} - n_{\text{actual}}|$. Multigenerational relative amplitude, modal alignment, and compatible share are $a_g = \|d_g\|/\|d_0\|$, $F_g = c_{\pm}(d_g, v)$, and $S_g = F_g^2$.

Carrier nomenclature. The archived R06/R07/C03/C07 code and CSV schemas use historical labels G0 and G1 for two operator carriers. To avoid collision with generation labels G1–G4, the manuscript calls these q_{cal} (calibration carrier) and q_{test} (test carrier). Archival filenames, JSON fields, and CSV column names are not rewritten.

1.3 Global evidence and inference audit

Table 2 exposes every experiment retained in the main paper or Supplement. The compact criterion column summarizes the frozen primary decision rules; the release also includes `provenance/EVIDENCE_AUDIT.csv`, which preserves the complete recovered H1–H4 strings from each available design JSON. Diagnostic and exploratory rows are not promoted to confirmatory status.

Table 2: Global audit of evidence status, design provenance, primary targets, frozen criteria, and outcomes.

ID	Status	Cohort	Design provenance	Primary estimand	Frozen primary criterion(s), compact form	Outcome
T01	confirmatory pass	73000–73063 (n=64)	original frozen JSON	response norm; response RSA	norm > 0.08; RSA shuffle advantages > 0; mean RSA > 0.07	PASS
T04	confirmatory fail/null	97000–97063 (n=64)	original frozen JSON	same-encounter gain / specificity	four signed predictions (< 0, < 0, > 0, > 0), each Holm/sign-flip + robustness	FAIL / null
T05	confirmatory pass	99000–99063 (n=64)	original frozen JSON	component norm and RSA	intact norm > 0.03; r ratio > 5; K RSA > 0.65; NO_R-intact RSA > 0.45	PASS
T06	confirmatory pass	101000–101063 (n=64)	original frozen JSON	interaction fraction; K/M relative RSA	interaction > 0.50; r ratio > 100; K/M RSA shuffle advantages > 0 with raw floors	PASS
T07	confirmatory pass	103000–103063 (n=64)	original frozen JSON	correct two-trace span R^2	pair-span R^2 > 0.90; correct–wrong > 0.25; pair–single > 0.08; older fraction > 0.30	PASS
F01	confirmatory pass	107000–107063 (n=64)	original frozen JSON	storage error vs readout nonadditivity	storage error < 0.02; response error < 0.03; readout excess > 0.05; mismatch < 0.02	PASS
F02	confirmatory pass	110000–110063 (n=64)	original frozen JSON	stagewise residual; additive- b collapse	row-norm residual > 0.04; centering gain > 0.02; sigmoid increment tiny; additive- b action residual < 10^{-4}	PASS
F03	confirmatory pass	113000–113063 (n=64)	original frozen JSON	third-trace contextual interaction	normalized b and action interactions > 0.10; additive- b removes > 0.999 / > 0.99	PASS
F04	confirmatory pass	115000–115063 (n=64)	original frozen JSON	jump/path fraction; event curvature	max jump/path < 0.10; rank and median curvature advantages > 0; median > generic rank	PASS
R01	confirmatory pass	121000–121063 (n=64)	original frozen JSON	cycle–flat response distance	cycle–flat response > 0.015; r > 0.010; m > 0.060; return mismatch > 0.020	PASS
R02	confirmatory pass	123000–123063 (n=64)	frozen working record; JSON unavailable	removed-response fractions	frozen ablation decomposition; exact standalone criterion file unavailable	PASS
R03	confirmatory pass	127000–127063 (n=64)	original frozen JSON	$x \rightarrow m$ and $m \rightarrow x$ relay	relay contrasts > 0.25, 0.20, 0.15, 0.15 with favorable-seed/bootstrap rules	PASS
R04	confirmatory pass	147000–147063 (n=64)	original frozen JSON	write/return alignment gains	return rescue > 0.005; specific rescue > 0.005; write cosine > 0.15; full return > 0.003	PASS
R05	confirmatory pass	149000–149063 (n=64)	original frozen JSON	projection rescue; G3 norm/cosine	G1 rescue > 0.08; G2 > 0.015; G3 norm < 0.015; G3 cosine > 0.55	PASS
R06	confirmatory pass	151000–151063 (n=64)	original frozen JSON	$\sigma_1(T)$; finite top-mode gain	mean $\sigma_1(T)$ < 0.15; finite top gain < 0.15; optimal–historical gain > 0.04	PASS
R07	confirmatory pass	153000–153063 (n=64)	original frozen JSON	$\sigma_1(A), \sigma_1(B)$; interface efficiency	$\sigma_1(A)$ < 0.15; $\sigma_1(B)/\sigma_1(A)$ > 5; ideal product < 0.20; efficiency < 0.75	PASS
M01	confirmatory pass	157000–157063 (n=64)	original frozen JSON	inherited vs reoriented gain	capacity change < 10%; G2 inherited alignment < 0.40; reoriented gain > 0.85; G4 reoriented > 0.70	PASS
M02	confirmatory pass	159000–159063 (n=64)	original frozen JSON	mode cosine/angle/projector drift	consecutive $ \cos $ > 0.995; angle < 2° ; projector < 0.05; relative σ_1 change < 0.05	PASS
M03	confirmatory pass	161000–161063 (n=64)	original frozen JSON	mode spread across histories	mean angle < 1.5° ; max angle < 3° ; σ_1 spread < 0.05; projector < 0.04	PASS
M04	confirmatory pass	163000–163063 (n=64)	original frozen JSON	alignment ² vs $\ \Delta m\ $	Spearman > 0.85; aligned/orthogonal imprint ratio > 2; positive contrast in > 95% seeds	PASS
M05	confirmatory pass	165000–165063 (n=64)	original frozen JSON	native/AMP/DIR compatibility	native ρ > 0.90; AMP/DIR ρ > 0.85; positive aligned contrasts in > 95% seeds	PASS
M06	confirmatory pass	168000–168063 (n=64)	original frozen JSON	operator-predicted imprint	norm correlation and vector cosine > 0.995; relative error < 0.005; shuffle advantages > 0.70	PASS
M07	diagnostic replication null	166100–166115 (n=16)	independent diagnostic record	native autocorrection effects	diagnostic replication only; no confirmatory threshold	diagnostic null
M08	confirmatory pass	170000–170063 (n=64)	original frozen JSON	compatible share in m and x	memory share > 0.75; returned share > 0.82; return increment > 0.04; residual sanity < 0.005	PASS
M09	confirmatory pass	172000–172063 (n=64)	original frozen JSON	$ u_1^T v_1 $; non-normality	$ u_1^T v_1 $ < 0.50; output- u_1 cosine > 0.995; preference > 0.40; σ_1/ρ > 1.5	PASS
C01	confirmatory equivalence/null	174000–174063 (n=64)	original frozen JSON	next-loop gain after co-product restoration	absolute gain differences < 0.002; ALL_NONX ratio in [0.98, 1.02]	NULL / equivalence
C02	diagnostic null	175000–175031 (n=32)	diagnostic working record	held-out common-map alignment	diagnostic held-out comparison; no frozen confirmatory threshold	diagnostic null
C03	confirmatory pass	176000–176063 (n=64)	original frozen JSON	own-instance transferred alignment	own $ \cos $ > 0.97; own–identity and own–wrong > 0.50; pair geometry > 0.98	PASS

ID	Status	Cohort	Design provenance	Primary estimand	Frozen primary criterion(s), compact form	Outcome
C04	confirmatory mixed/fail	180000–180063 (n=64)	transcript-recovered design	reciprocity-specific closure gains	reciprocal advantage over matched controls > 0.05/0.04/0.025; vs native > 0.04	mixed; strong H fail
C05	confirmatory fail/null	182000–182063 (n=64)	transcript-recovered design	subspace/basis closure gains	XSPACE gain > 0.08; memory-only weak; interaction > 0.08; BOTH alignment > 0.55	FAIL / heterogeneous
C06	exploratory null	same 182000–182063 cohort (n=64 reused)	transcript-recovered design	static moderators; CV R^2	post-confirmatory exploratory moderator predictions; no confirmatory threshold	exploratory null
C07	confirmatory pass	184000–184063 (n=64)	original frozen JSON	dynamic alignment; closure-need MAE	dynamic $ \cos > 0.995$; vs static > 0.20; vs wrong > 0.70; need MAE < 0.02	PASS
G01	confirmatory pass	186000–186063 (n=64)	original frozen JSON	G4 amplitude; specificity; alignment	G4 > 0.85; vs unclosed > 0.70; specificity > 0.60; mean G1–G4 alignment > 0.99	PASS
G02	confirmatory pass	188000–188063 (n=64)	original frozen JSON	G4 share; 67.5°/90° boundary	nonorthogonal G4 share > 0.95; 67.5° gain > 0.70; vs unclosed > 0.65; G1 90° < 0.10	PASS

Inference was experiment specific. Confirmatory rows used the released frozen H1–H4 rules, generally combining effect-size thresholds, favorable-seed or favorable-cell fractions, bootstrap intervals, and, where specified in the design artifact, one-sided sign-flip/randomization tests with Holm correction. The exact design files and the exhaustive machine-readable audit should be treated as authoritative when a compact table entry omits a secondary threshold.

1.4 Trace-weight continuation

The main paper identifies median-relative collective readout as the principal source of nonadditivity in the K branch. F04 varied trace weights continuously while following b and downstream action. Generic rank-order events were distinguished from events in which a row crossed the collective median reference. The primary observables included maximum local jump relative to total path length and curvature around rank and median events. This experiment refined the mechanistic interpretation rather than serving as a necessary link in the main causal chain.

1.5 Turning-point support ablations

R02 selectively reset fast and slow state families at the turning point of a path-dependent trajectory and quantified the loss of later historical response. These intervention fractions are protocol specific and are not information-theoretic decompositions. The experiment motivated the subsequent explicit focus on x and m .

1.6 Partial return-geometry interventions

R04 tested whether more limited external alignment of write or return geometry could improve relay performance before the complete dynamic closure was constructed. Return geometry and write orientation were manipulated externally and compared with native and scrambled controls. The experiment served as a bridge between causal reentry and the later separation of geometry from gain.

1.7 Modal stationarity across transplanted history fields

M03 retained a common fast/current carrier while transplanting complete slow historical packages $[m, \theta, r, K, b, g_h]$ from eight histories. The local loop operator was estimated under identical conditions after each transplant. Pairwise dominant-mode angles, maximum within-seed displacement, singular-gain spread, and projector distance measured the sensitivity of the transmission geometry to historical state.

1.8 Compatibility-imprint decomposition

M04 introduced equal-norm recurrent perturbations spanning different alignments with the dominant input mode and measured slow-memory displacement after a short neutral sensing interval. M05 transplanted manipulated versions of this imprint into a common carrier. `AMP_ONLY` preserved compatibility-dependent imprint magnitude while controlling direction; `DIR_ONLY` preserved direction while controlling magnitude. These experiments test internal availability of compatibility information, not endogenous use of that information for correction.

1.9 Alternative closure hypotheses

C02–C06 tested progressively stronger static alternatives. C02 fitted common cross-instance maps (identity, orthogonal Procrustes, ridge, and a tested state-conditioned reader). C03 tested a simple within-instance orthogonal bridge across contexts. C04 altered nominal write/read architecture to test scale-invariant reciprocity. C05 independently manipulated recurrent-state

subspace and memory-basis alignment while preserving singular values. C06 used the same cohort exploratorily to test whether simple static mismatch descriptors predicted heterogeneous benefits. These blocks form a falsification path toward the effective causal construction C07.

1.10 Post-confirmatory robustness audits

The following audits were added during adversarial pre-release review. They were not part of the prospectively frozen confirmatory criteria and therefore do not alter the pass/fail status of G01, C07, or the other frozen blocks. They use the released portable/reference-reconstruction pathway and are reported as robustness analyses only.

Closure renormalization audit. The G01 implementation applies C_{dyn} to the returned packet and then restores its pre-closure norm so that the closure operation itself contributes direction but not magnitude. On the 64-seed portable reference reproduction, the required multiplicative renormalization factor was extremely close to one across all four generations: the overall mean was 1.000243, the median 1.000099, and the observed range across 256 seed-generation packets was 1.000005–1.005344. In an otherwise identical post-hoc `FIXED_DYNAMIC_NO_RENORM` branch, generation-4 mean relative norm was 0.98163, versus 0.98255 with norm restoration in the same reference reproduction; generation-4 directional alignment differed by at most 2.3×10^{-5} across seeds. The archived confirmatory G01 value remains the source of record; this audit shows only that the explicit post-closure norm restoration is not materially responsible for the reproduced persistence.

Multiple context-pair transfer audit. The original C07 confirmatory design froze one calibration-to-test pair. We subsequently repeated the same construction on four symmetry-related pairs drawn from the four relation triples

$$(-135, -45, 45), (-45, 45, 135), (45, 135, -135), (135, -135, -45),$$

using generation 0 for calibration and generation 1 for testing, on the same 64-seed cohort. The four transfers were pair $0 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 0$, and $3 \rightarrow 1$. Across 256 seed-pair cells, mean own-instance dynamic alignment was 0.999576 and 254/256 cells exceeded 0.99. Mean advantage over the tested static map was 0.416856, positive in all 256 cells. Closure-need mean absolute error was 0.004611, with 253/256 cells below 0.03. Because the pairs were selected after the confirmatory study and reuse its cohort, this is robustness evidence rather than a new confirmatory generalization claim.

Finite-difference sensitivity audit. On seeds 184000–184015, C07 was re-estimated with central-difference steps $\epsilon = 10^{-3}, 10^{-4}, 10^{-5}$. Mean transferred alignment was 0.999748 at all three scales to the displayed precision. The leading input/output modal vectors at 10^{-3} and 10^{-5} had sign-invariant cosines greater than 0.999999999999998 with the 10^{-4} baseline across the audited seeds; relative changes in the calibration leading singular value were below 10^{-7} on average. This supports numerical stability with respect to the chosen finite-difference scale over the tested decade range. It does not establish invariance to the 24-step operator window, which remains part of the operational definition.

The released files `results/robustness/` and `code/posthoc_audits/` contain the corresponding tables and executable audit scripts.

2 Supplementary results

2.1 A maladaptive second-encounter interpretation was not supported

An early hypothesis proposed that a historical perturbation might make a second encounter with the same relation systematically maladaptive. T04 tested this prediction with common-present controls, same and orthogonal second perturbations, and relevance/plasticity ablations. The intact common-present same-relation effect was $+0.001772$, opposite to the predicted maladaptive direction, and the planned relevance-removal rescue was not observed. The result pruned a valence-based interpretation of the historical trace without undermining later path-dependence results.

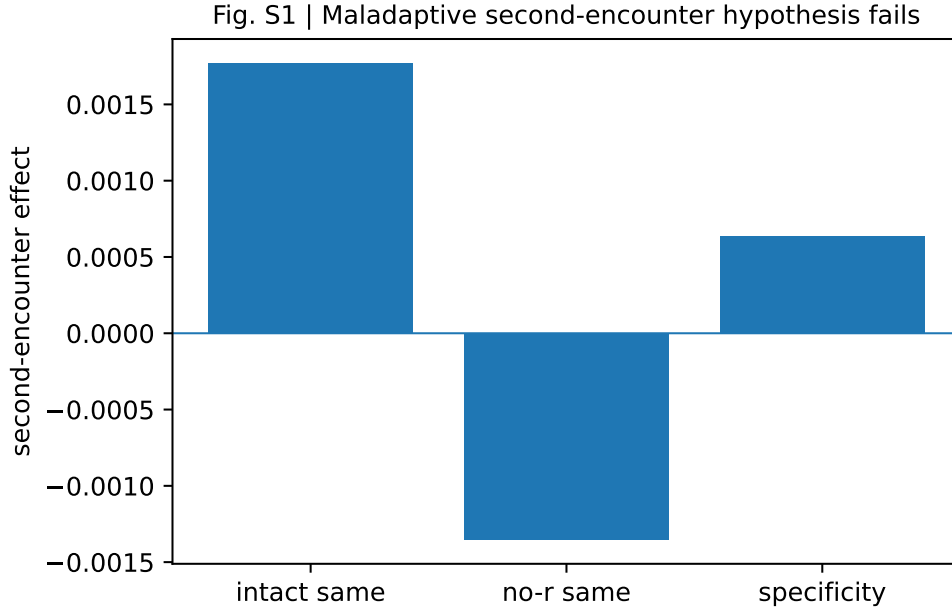


Figure 1: The maladaptive second-encounter hypothesis fails. The intact same-relation effect is small and positive rather than maladaptive; relevance removal does not produce the planned rescue.

2.2 Collective readout is structured but continuous

F04 showed that contextual reweighting was not dominated by a single catastrophic discontinuity. The maximum-jump-to-path ratio was approximately 0.0741 for b and 0.0771 for action. Curvature increased more strongly around median crossings than around generic rank events; for b , the mean median-over-rank curvature advantage was 2.37×10^{-4} . Thus the collective median acts as a structured continuous nonlinearity rather than merely as a binary switch.

2.3 Both active and slow supports carry turning-point history

Removing the slow-state family at the turn eliminated mean fraction 0.538682 of the later response. Within that slow contribution, resetting m alone removed 0.476297, corresponding to 88.67% of the slow-family effect. Removing fast state accounted for fraction 0.341911, and resetting x alone removed 0.290799, corresponding to 84.17% of the fast-family effect. These results motivated the explicit $x \leftrightarrow m$ relay studied later.

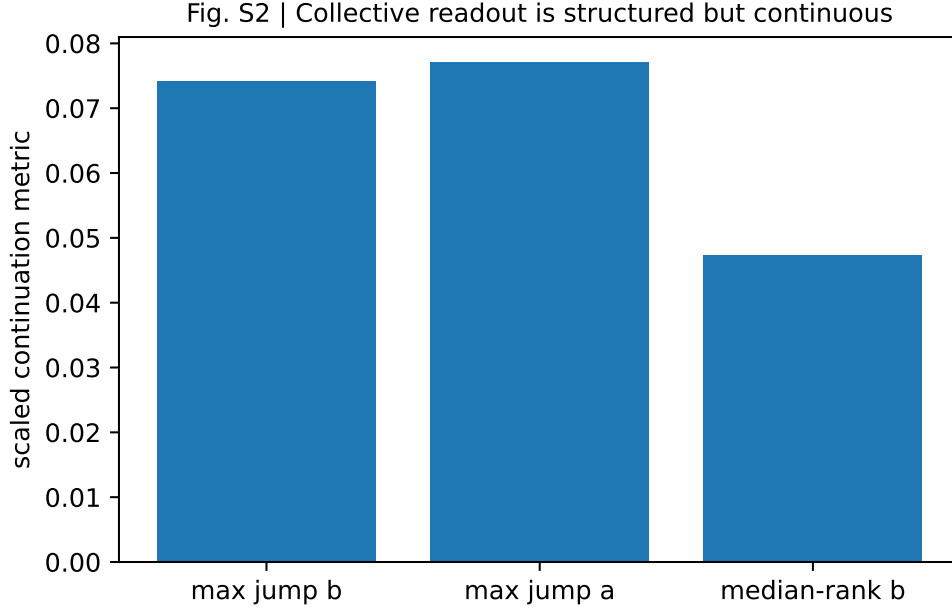


Figure 2: Continuation analysis of collective trace readout. Maximum local jumps remain small relative to full path length; median crossings produce stronger curvature than generic rank events. The median-minus-rank b advantage is multiplied by 200 only for the display scale.

2.4 Partial geometric alignment improves but does not close the relay

Return alignment increased recurrent-state target projection by $+0.009367$ relative to native and by $+0.012342$ relative to scrambled return. Write-side alignment increased the relevant memory-direction cosine by $+0.184014$ and increased the full-return target projection by $+0.004269$. These improvements established a measurable role for geometry but were far too small to prevent multigenerational extinction.

2.5 Historical fields do not substantially relocate the dominant loop mode

Across eight transplanted slow historical packages, the mean pairwise dominant-mode angle was only 0.7953° , the mean maximum within-seed angle was 1.6135° , mean projector distance was 0.01963 , and the leading singular-gain spread was 1.13% . Historical state can therefore alter downstream operation without substantially relocating the dominant transmission geometry in this protocol.

2.6 Compatibility is multiplexed in memory amplitude and direction

In M04, compatibility with the dominant input mode strongly predicted slow-memory imprint magnitude: mean Spearman $\rho = 0.926933$, with aligned-to-orthogonal imprint ratio 2.43214 . M05 then separated magnitude and direction. The native relation was $\rho = 0.971269$; preserving only the compatibility-dependent imprint amplitude retained $\rho = 0.923763$, while preserving only imprint direction retained $\rho = 0.862314$. Both channels therefore carry substantial compatibility information.

2.7 Closure is not captured by a common cross-instance map

The diagnostic C02 common-map analysis yielded held-out mean absolute alignment 0.230768 for identity, 0.127457 for common Procrustes, 0.160252 for ridge regression, and 0.118396 for the

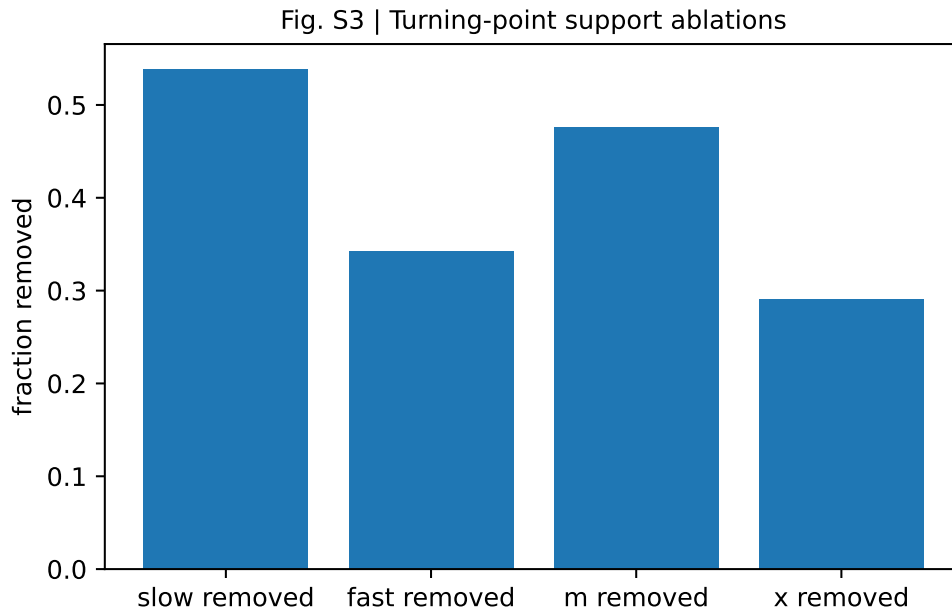


Figure 3: Turning-point support decomposition. Both slow and fast families contribute, with m and x dominating their respective families.

tested state-conditioned reader. No tested common map improved on identity. This diagnostic does not rule out all possible universal nonlinear transformations.

C03 asked a narrower question. A simple within-instance orthogonal bridge reconstructed in calibration context transferred to a distinct context of the same reactor with mean absolute alignment 0.996353. It beat identity by 0.697708 and a wrong-instance map by 0.707588, with all confirmatory seeds favorable. Input and output modal frames were themselves highly stable across contexts (0.997906 and 0.998170 mean alignment). These observations motivated the instance-specific effective causal construction used in C07.

2.8 Static architectural explanations are insufficient

C04 tested the simplest reciprocity hypothesis. Exact reciprocity improved one descriptive comparison over native by +0.18397, but the stronger frozen reciprocity-specific hypotheses failed: their primary mean advantages were +0.07051, +0.02441, and +0.01429, with favorable-seed fractions 0.6875, 0.59375, and 0.546875. Reciprocity altered behavior but did not robustly supply the predicted closure.

C05 independently manipulated recurrent-state subspace alignment and memory-basis alignment. Recurrent-space matching produced mean effect +0.10107 but only 73.44% positive seeds. Memory-basis matching alone produced mean signed effect -0.01849 . The interaction was +0.08755 with only 68.75% favorable seeds, and matching both yielded mean input-output alignment 0.52495. All frozen primary hypotheses failed. The exploratory C06 moderator analysis also found no useful simple static predictor of heterogeneity; representative correlations were Pearson $r = -0.159$ and Spearman $\rho = -0.107$, while cross-validated ridge models produced negative R^2 values (-0.037 and -0.085).

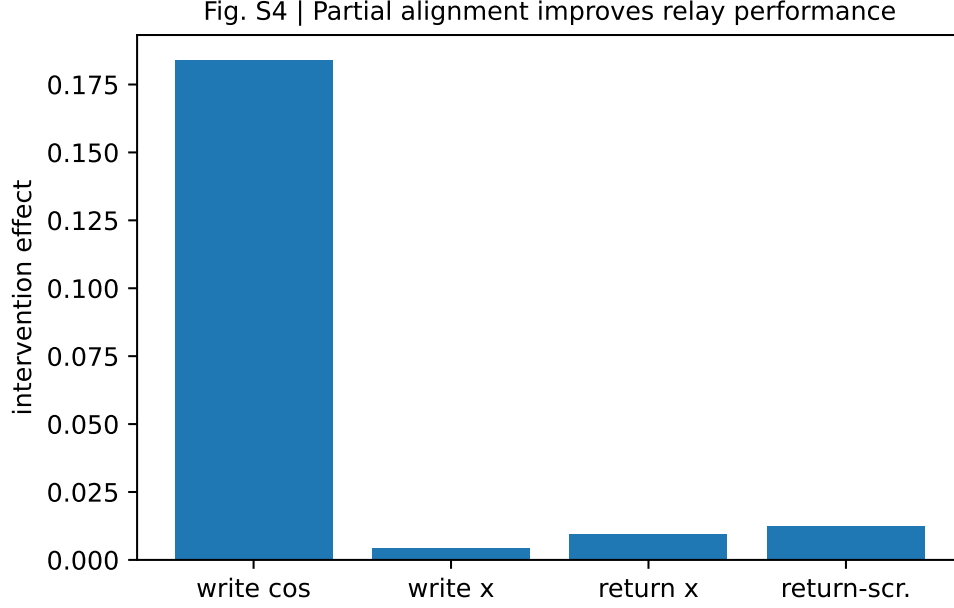


Figure 4: Partial write- and return-side alignment interventions improve relay performance but remain insufficient for repeated recurrence.

3 Confirmatory criteria for the final repair

Table 3: Frozen G01 and G02 criteria and observed outcomes.					
Test	Frozen rule	Outcome			Pass
G01 H1	G4 mean > 0.85; > 90% seeds > 0.70; bootstrap lower bound > 0.75	mean	0.992131,	fraction	yes
			0.96875,	CI	
			[0.960493, 1.021435]		
G01 H2	correct closure advantage over same-gain/no-closure > 0.70; > 95% positive; lower bound > 0.60	+0.912540,	all positive,		yes
		CI [0.881205, 0.941729]			
G01 H3	advantage over better STATIC/WRONG control > 0.60; > 90% positive; lower bound > 0.50	+0.792420,	all positive,		yes
		CI [0.756705, 0.827210]			
G01 H4	mean G1–G4 postclosure alignment > 0.99; > 90% seed minima > 0.98	mean	0.999136,	fraction	yes
		1.0			
G02 H1	G4 nonorthogonal mean share > 0.95; > 90% cells > 0.90; lower bound > 0.93	0.990252,	fraction		yes
			0.98958,	CI	
		[0.987357, 0.992740]			
G02 H2	67.5° increase > 0.70; > 90% positive; lower bound > 0.65	+0.833120,	all positive,		yes
		CI [0.825598, 0.839580]			
G02 H3	FIXED minus same-gain/no-closure > 0.65; > 95% positive; lower bound > 0.60	+0.873150,	all positive,		yes
		CI [0.860523, 0.885624]			
G02 H4	$S_{90^\circ, G1} < 0.10$, $S_{67.5^\circ, G1} > 0.75$, and > 85% seeds with $S_{67.5} > S_{90}$	0.001544 vs 0.864123, all seeds ordered			yes

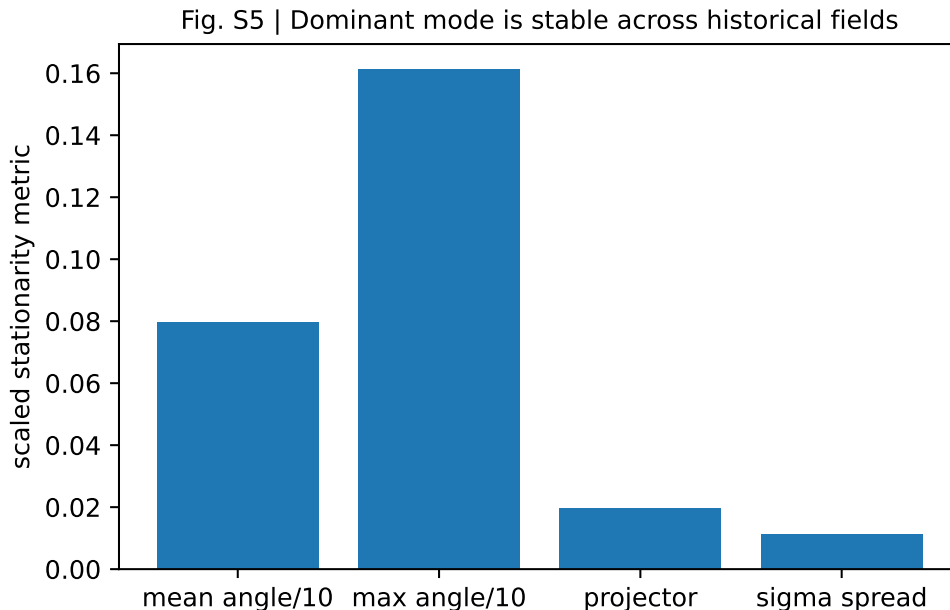


Figure 5: The dominant loop mode is stable across transplanted historical fields. Angle values are divided by ten only for the shared display scale.

4 Reproducibility and provenance

The public release distinguishes three provenance classes. **ORIGINAL** denotes artifacts recovered independently from the persistent Library or working archive. **TRANSCRIPT_RECOVERED** denotes exact code or design text copied from visible code/JSON blocks in the frozen research transcript. **REFERENCE_RECONSTRUCTION** denotes executable replacements reconstructed from the frozen record, recovered designs, original base reactor, downstream APIs, and archived result schemas.

Three transient dependencies fall in the third class: the relational $R_4(\phi)$ scaffold, the full-loop operator scaffold, and the half-channel factorization scaffold. The reconstructed relation passes its prospectively specified causal-signature test across $-135^\circ, -90^\circ, -45^\circ, 0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ$. The two reconstructed operator scaffolds were audited against archived confirmatory tables. For the eight seed-level validation cases, primary singular-value quantities show correlations approximately 0.98–0.999 with low mean absolute error. Running the reconstructed operators on the full frozen cohorts preserves the primary R06 and R07 confirmatory decisions.

The original transient noise-seed mixer was not independently archived. The reconstruction preserves matched-noise semantics but cannot claim forensic equality of every auxiliary seedwise column. The release therefore claims an executable, source-auditable scientific reproduction pathway, not byte-identical recovery of every transient original file.

Raw research-conversation exports are not distributed in the public repository because they contain unrelated material. Their filenames and SHA-256 hashes are recorded in the provenance manifest, while the scientific design and result artifacts extracted from them are released separately.

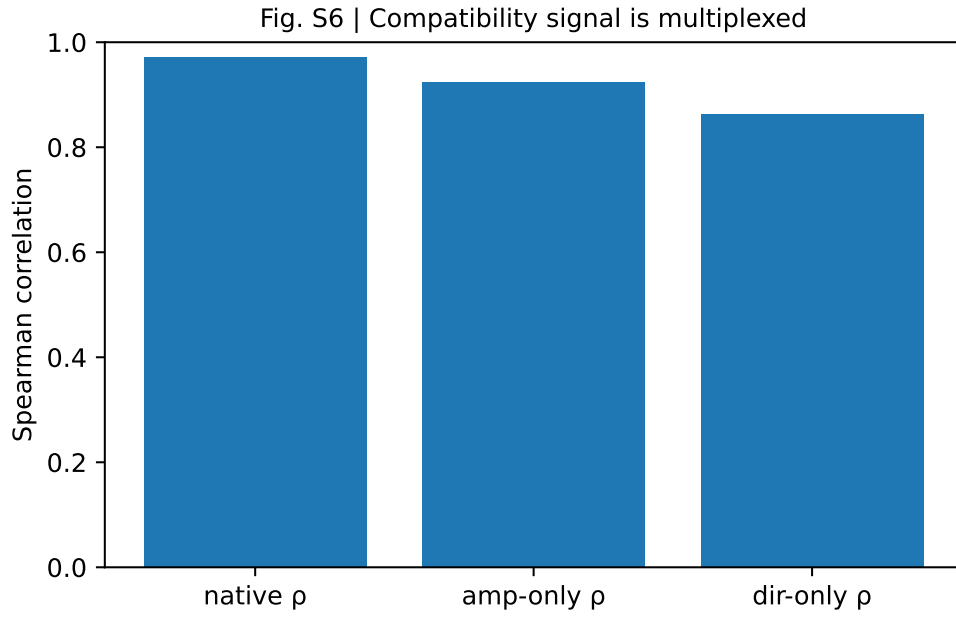


Figure 6: Compatibility information is multiplexed in slow-memory amplitude and direction.

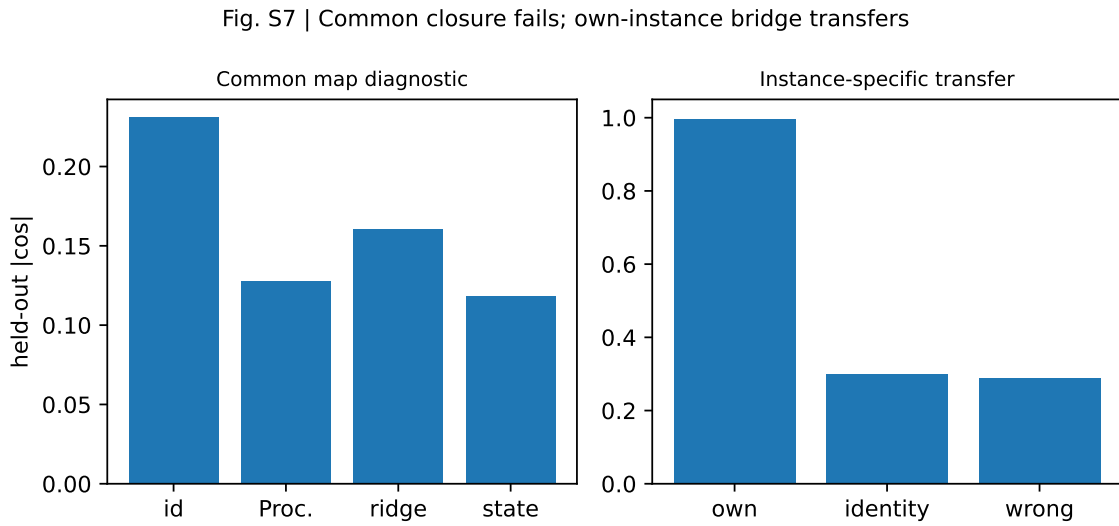


Figure 7: Common closure maps fail in the tested diagnostic, whereas a simple own-instance bridge transfers across the tested context change.

Fig. S8 | Static architectural explanations are insufficient

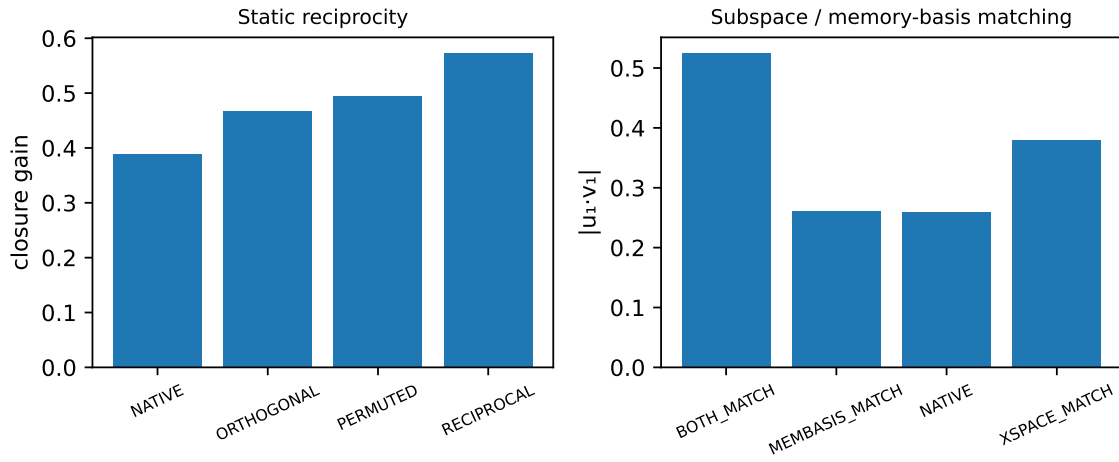


Figure 8: Static architectural explanations are insufficient. Reciprocity and subspace/basis matching alter the geometry but do not satisfy the stronger frozen closure predictions.